

METHOD,URL and Status codes

▼ General	
Request URL	https://www.wikipedia.org/
Request Method	GET
Status Code	● 304 Not Modified
Remote Address	[2001:df2:e500:ed1a::1]:443
Referrer Policy	strict-origin-when-cross-origin

Response Headers:

▼ Response headers	
Accept-Ranges	bytes
Age	49971
Cache-Control	s-maxage=86400, must-revalidate, max-age=3600
Content-Encoding	gzip
Content-Type	text/html

Age-indicates the time in seconds for which an object was in a proxy cache.

Content-Type-Tells the browser what type of content was returned.

Content-Encoding-Shows how the response was compressed, if applicable.

Cache-Control-Specifies caching rules for the response.

Request Headers:

Accept	text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding	gzip, deflate, br, zstd
Accept-Language	en-US,en;q=0.9
Cache-Control	max-age=0

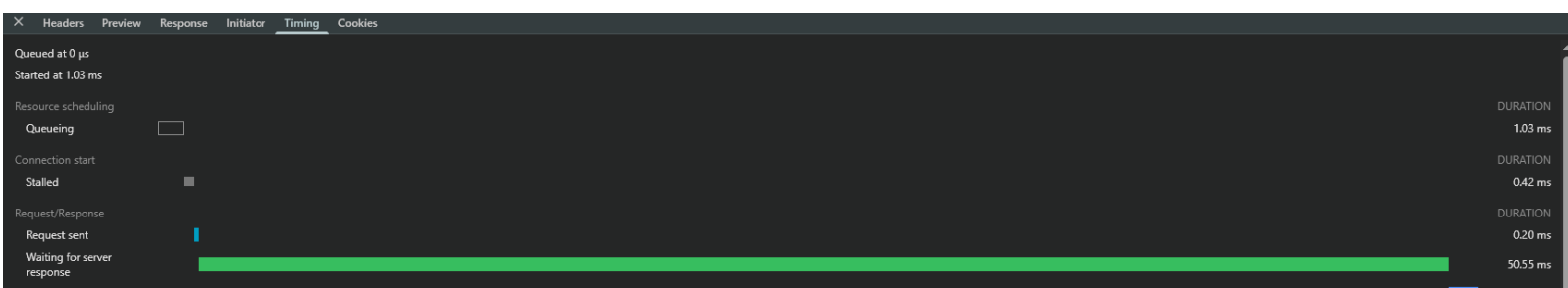
Accept : Request -Indicates accepted content types.

Accept-Encoding :Request -Indicates supported compression formats.

Accept-Language :Request -Indicates preferred languages.

Cache-Control :Response -Provides caching instructions.

Timing Breakdown:



How everything works together?

DNS first converts the domain name of the website into an IP address. The browser then makes the appropriate network connection with the server. TLS negotiation occurs to create a secure connection because the chosen website uses HTTPS. The browser sends an HTTP request, typically a GET request for the primary document, after establishing a secure connection. The requested resource and a status code are included in the HTTP response that the server sends back. The response is then downloaded and processed by the browser.

Difference b/w HTTP and HTTPS:

Although HTTP does not offer encryption on its own, it does specify how clients and servers exchange online resources. TLS and HTTP are combined in HTTPS. By encrypting the transmission and offering server authentication and integrity protection, TLS safeguards HTTP communication. An SSL/TLS stage that would not exist as an HTTP application-layer security step on a plain HTTP connection is seen in the DevTools timing information for the HTTPS request.

Screenshot of network tab:

